MANEL MORSLI

Intelligent Systems Engineer | AI Engineer | Full-Stack AI Developer

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Portfolio · GitHub · LinkedIn

PERSONAL PROFILE

Full-Stack AI Engineer and Product Owner with end-to-end experience delivering intelligent platforms—from writing functional specifications and designing system architecture, to building production backends, admin dashboards, and public-facing AI widgets. Skilled in bridging the gap between technical development and product strategy: planning roadmaps, producing release notes, and driving features from concept to v0.0.1 and beyond. Passionate about AI, knowledge-based systems, object detection, image processing, and scalable RAG pipelines. Currently seeking a mentorship relationship with an experienced AI project manager or technical lead to deepen expertise in AI product management, architectural decision-making, and strategic skill-scaling—with the goal of contributing to high-impact, research-driven AI initiatives at an international level.

TECHNICAL SKILLS

Languages: Python, C / C++, Matlab, JavaScript / TypeScript, PHP, SQL, YAML, LaTeX, HTML / CSS, Linux shell

AI / ML / NLP: PyTorch, TensorFlow, Transformers (HuggingFace), Sentence-Transformers, Jina-Embeddings-v3 (late chunking), fastembed, FlashRank (reranking), LangChain, LlamaIndex, YOLO, SAM (Segment Anything), Scikit-learn, Generative AI, Prompt Engineering

Computer Vision: YOLO v8/v11, SAM, Roboflow, OpenCV, Data Annotation (JSON / YAML)

Backend & APIs: FastAPI, SQLAlchemy (async), Alembic, Pydantic, JWT / OAuth2, Taskiq (async task queue), asyncpg, REST APIs

Databases & Storage: PostgreSQL + pgvector, Redis, SeaweedFS (S3-compatible), InfluxDB, MySQL, Vector databases

Web Scraping & Crawling: Playwright, Crawlee, Crawl4AI, Trafilatura, Selectolax, BeautifulSoup

Frontend: React, Next.js, Vite, Tailwind CSS, HTML5 / CSS3

DevOps & Prod infrastructure: Docker, Docker Compose, Git / GitHub, CI/CD basics,

Knowledge Representation: Ontologies, OWL, RDF / RDFS, Protégé

IoT & Embedded: Arduino, Node-RED, Sensor networks, Grafana, OpenWeatherMap API

Product & Project Management: Product Ownership (PO), Cahier des charges / Functional specifications, System & solution architecture design, Release management & release notes, Technical reporting, Roadmap planning, User story writing, Knowledge base management, Agile / Scrum

Methodology: End-to-end AI product delivery, Data mining, System architecture design, Technical documentation, API design

TECHNICAL PROJECTS

AI-Powered Multi-Tenant RAG Platform & Knowledge Management System — December 2025 – Current

- Designed and built a production-grade, multi-tenant with data isolation Retrieval-Augmented Generation (RAG) platform from scratch, supporting organisations, workspaces, documents, and conversational AI.
- Backend: FastAPI + SQLAlchemy (async) + Alembic migrations + JWT authentication; async task processing via Taskiq / Redis workers; Playwright & Crawlee for intelligent web crawling and document ingestion.
- Vector search: PostgreSQL + pgvector; local LLM inference via Ollama (Qwen 2.5, Mistral, BGE-M3); FlashRank cross-encoder reranking pipeline.
- Storage layer: SeaweedFS S3-compatible object store for PDFs, CSVs, and binary assets.
- Research module: late-chunking experiments with Jina-Embeddings-v3 (sentence-transformers), PyMuPDF4LLM parsing, and a React / Vite research UI for embedding strategy evaluation.
- LangChain & LlamaIndex integrated for chain-of-thought orchestration and retrieval pipelines.
- Full Docker Compose orchestration (backend, frontend, Next.js dashboard, Ollama, Redis, PostgreSQL, SeaweedFS) for one-command local deployment.
- Stack: Python 3.12 · FastAPI · PostgreSQL · pgvector · Redis · Ollama · LangChain · LlamaIndex · Docker · React · Next.js · SeaweedFS

Master's Thesis — Intelligent Early-Diagnosis System for Breast Cancer via TCM — UMMTO | 2024 – 2025

- Designed an AI-driven diagnostic pipeline combining Traditional Chinese Medicine tongue analysis with modern computer vision for breast-cancer early detection.
- Used YOLO v8 for object detection, SAM for segmentation, Roboflow for dataset management, and Generative AI for synthetic data augmentation.
- Built end-to-end ML pipeline: data annotation → training → evaluation → inference API.

University Chatbot — Student FAQ System — Academic UMMTO | 2024

- Developed an NLP chatbot for student FAQ automation using fine-tuned GPT and T5 models (HuggingFace Transformers).
- Implemented intent classification, retrieval-based answering, and fallback generation.

Ontology Project — Fashion Industry Knowledge Graph — Academic UMMTO | 2024

- Built an interactive fashion-industry ontology using Protégé, OWL, RDF, and RDFS.
- Defined class hierarchies, object/data properties, and SPARQL query interfaces.

Data Mining Benchmark Suite — Academic UMMTO | 2023

- Compared regression, clustering (K-Means, DBSCAN), and classification (SVM, Random Forest, KNN) on multiple real-world datasets; visualised results with Matplotlib / Seaborn.

IoT Project — "Smart Greenhouse" Real-Time Monitoring System — Academic | 2023

- Built a sensor-driven greenhouse monitoring system using Arduino, Node-RED, InfluxDB, OpenWeatherMap API, and Grafana dashboards for real-time alerting.

Bachelor's Final Project — "DentalPlan" Web Application — UMMTO, Team Lead | 2021-2022

- Led a team to deliver a dental appointment-booking web app (HTML, CSS, JavaScript, PHP, MySQL).

Startup Concept — Platform for Artisans & Manual Workers — Personal | 2022

- Proposed, modelled, and pitched a marketplace startup concept connecting skilled artisans with customers; produced business model canvas and technical architecture.

PROFESSIONAL EXPERIENCE

Full Stack AI Engineer & Product Owner — AI Chatbot Platform — High Commission of Digitalization Algeria | December 2025 – Current

- Owned the product from day one: wrote the full functional & technical specifications, defined the system architecture, produced the solution design document, and authored all project reports and release notes up to v0.0.1.
- Acted as Product Owner: built and maintained the product backlog, prioritised features, defined acceptance criteria, and planned each delivery milestone to ensure a coherent, scalable platform.
- Designed and developed the Admin Dashboard — a Next.js back-office enabling administrators to manage the knowledge base (documents, workspaces, tags, ingestion jobs) and configure the AI behaviour that powers the public-facing chat widget.
- Built the embeddable public chat widget (React + Vite) consumed by end-users, backed by a FastAPI RAG pipeline (pgvector semantic search + Ollama local LLM inference).
- Architected multi-tenant data isolation (organisations → workspaces → documents) with JWT authentication, async task workers (Taskiq / Redis), and S3-compatible file storage.
- Delivered end-to-end: requirements → architecture → development → testing → release notes → versioned deployment via Docker Compose.
- Stack: Python · FastAPI · PostgreSQL · pgvector · Redis · Ollama · LangChain · LlamaIndex · React · Next.js · Docker · SeaweedFS

Computer Science Teacher — Middle School, Algeria | January 2025 – May 2025

- Delivered computer-science curriculum to middle-school students.
- Developed strong communication, classroom management, and lesson-planning skills.

IT & Telecom Intern — Banque d'Extérieure d'Algérie, Algeria | December 2024

- Gained overview of banking operations; explored IT project management and telecom division (datacenter administration and data transmission).

IT Intern — Regional Directorate Hassi Messaoud DP, Sonatrach, Algeria | July – August 2024

- Explored IT departments, industrial development, database management, networking & security services, and datacenter operations at a major energy company.

EDUCATION

Master's Degree — Intelligent Computer Systems — Mouloud Mammeri University (UMMTO), Tizi-Ouzou, Algeria | 2022 – 2025

- Specialisation in Artificial Intelligence — Machine Learning, Deep Learning, NLP, Computer Vision, Knowledge Representation, Distributed Systems.

Bachelor's Degree — Computer Systems — Mouloud Mammeri University (UMMTO), Tizi-Ouzou, Algeria | 2019 – 2022

- Strong foundation in computer systems, algorithms, databases, and software engineering.

Baccalaureate — Mathematics — Aghri Mohammed Said High School, Azeffoun, Tizi-Ouzou, Algeria | 2019

LANGUAGES

Tamazight (Native) · **Arabic** (Fluent) · **French** (Professional Proficiency) · **English** (Professional Proficiency)

CERTIFICATES & TRAINING

- Computer Maintenance & Network Installation — Tizi-Ouzou, Algeria
- Self-directed study: Docker, DevOps, Cloud Computing, Cybersecurity fundamentals (ongoing)

INTERESTS

Artificial Intelligence · Quantum Computing · Medical Technology · Cybersecurity · DevOps & Cloud · Scientific Research · Mathematics · Teaching · Entrepreneurship & Business · Fashion Design · Traditional Sewing · Traditional Medicine · Creative & Innovative Tech Solutions